

SPEECH PROSODY: A METHODOLOGICAL REVIEW

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This critical review is mainly concerned with methodological issues in prosody research, with the aim to highlight progress toward developing predictive knowledge about prosody. The review shows that there has been a steady progression in terms of methodological rigor as the field goes through major methodological trends that can be described as analysis-by-introspection, analysis-by-transcription, analysis-by-hypothesis-testing and analysis-by-modeling. All the major methodologies currently still co-exist and each still has its own merit. But all of them are evaluated in terms of their effectiveness in establishing knowledge that is generalizable. Finally, an emphasis will be made on the need to have much more linking and integration between different subareas of prosody research.

Keywords: analysis-by-introspection; analysis-by-transcription; analysis-by-hypothesis-testing; analysis-by-modeling; predictive knowledge; degrees of freedom.

1 Introduction

Speech conveys information not only through segmental sounds like vowels and consonants, but also through prosody, i.e., variations in fundamental frequency (also referred to as F_0 and pitch), duration, intensity and voice quality. Unlike in the case of segments, one of the greatest difficulties about prosody is what I would call the *lack of reference problem*. By reference I mean a pivot that serves as both a starting point of inquest and a point that one can comfortably fall back on. In segmental research, for example, word identity serves as such a reference, because it is consciously accessible whether the language under study has a writing system or even whether the human informant is literate. Thus we can confidently investigate the phonetic properties that distinguish one word from another. But words also give us a false sense of certainty, because their ease of access may lead to the assumption that what underlies the lexical contrast, namely, the phonemes, are also easily accessible to individual speakers. This has been shown not to be the case by the discovery of phonological awareness (Mattingly, 1972; Liberman et al. 1990). That is, awareness of the segments needs to be either introduced or enhanced through literacy education (Bentin et al., 1992; Mann & Brady, 1988). So, the existence of orthographic representations of a functionality does not guarantee its conscious awareness and hence easy observation. In the case of prosody, very little of its functionality is orthographically represented, except for the punctuations whose meanings are at best ambiguous. Thus the starting point of inquest of prosody is inevitably vague and arbitrary, and it is just as difficult to know for certain what to check against after an observation is made. The difficulty is further compounded by the fact that our pitch awareness is not nearly as high as is often assumed, especially when it comes to melodic events in prosody (Dankovicova et al., 2007). As a result, the identification of the prosodic functionalities and their phonetic realizations often has to be attempted at the same time, with limited help from our built-in introspective ability. This is probably why approaches to prosody have been so disparate. There is therefore an urgent need to examine prosody research from the perspective of methodology. This paper offers a brief review of the strategies that have been attempted over the last century or so, with the hope that this may help future research to achieve greater efficiency in finding real solutions.

There have been a number of reviews of prosody research, the most recent only a year ago (Wagner & Watson, 2010). The emphases of those reviews are different from what is covered here. Shattuck-Hufnagel and Turk (1996) and Wagner and Watson (2010) were both concerned mainly with prosodic structures as reflected by prominence and boundary strength and their phonological representations. Cutler et al. (1997) focused on whether and how prosody may help the perceptual processing of words, syntactic structures and discourse structures. Botinis et al. (2001) was quite comprehensive and covered some of the issues discussed in the current review. But it was meant to be a tutorial and hence was in general fairly noncritical. Also much has happened in the past 10 years in terms of new empirical data and theoretical development. The current review will focus on the progress over the years toward solving the lack of reference problem by trying to build up predictive rather than descriptive knowledge about prosody.

2 Analysis by transcription

Systematic examination of prosody can go back as early as Walker (1787), who proposed a tone marking system for English intonation which is not very different from the IPA annotations for lexical tones. The modern British intonation tradition is a continuation of this approach, with

representative work by Palmer (1922), Kingdon (1958), O'Connor and Arnold (1961), Halliday (1967), Crystal (1969), Cruttenden (1997) and Wells (2006). In this tradition, intonation is portrayed by a transcription system consisting of representations for prominences (usually by the size of successive dots corresponding to the stressed syllables) and contours (by curved lines, sometimes with arrow heads to indicate the direction of pitch movements). Parallel to this tradition, there have been transcription systems in America that put greater emphasis on tonal levels rather than tonal contours. This tradition can go back to Rush (1827), and the subsequent work includes Pike (1945), Trager and Smith (1951), Hockett (1958) and Pierrehumbert (1980). One exception from this tradition is Bolinger (1986, 1989), who used a transcription system that represents pitch contours by word-art like text arrangements. Despite the differences in the proclaimed general goals (one mainly for teaching and the other for analysis), the two traditions share one thing in common, namely, they both try to understand intonation through *analysis-by-transcription*. This is despite the improvement over the years on the tools of transcript from primarily auditory detection to instrumental observation. The latest major development of this approach, which is still widely practiced, is the transcription system known as ToBI for TOnes and Break Indices (Silverman et al., 1992). The system is developed based on the pitch accent representations proposed by Pierrehumbert (1980) and the boundary representations proposed by Price et al. (1991).

The methodological motivation behind the analysis-by-transcription approach is probably well characterized by Shattuck-Hufnagel & Turk (1996:193): "Because current theories do not predict the precise prosodic shape that a particular utterance will take, it is important to determine the prosodic choices that a speaker has made for utterances that are used in an auditory sentence processing study." In other words, given the fact that our knowledge about prosody has not yet developed to the level of being able to make precise predictions, there still needs to be means of referring to whatever prosodic forms we observe. It is interesting to note, however, the discrepancy between this characterization and the assumption widely held within this approach that the transcribed categories such as pitch accents, boundary tones and phrase tones are *phonological* rather than *phonetic*, i.e., the transcriptions represent phoneme-like rather than allophone-like units. As explained in Pierrehumbert (1980:59), the establishment of these categories was not based on whether or not they could clearly distinguish meanings, but rather was the outcome of "attempting to deduce a system of phonological representation for intonation from observed features of F_0 contours." In an effort to link the form-based intonation categories proposed in Pierrehumbert (1980), Pierrehumbert & Hirschberg (1990) proposed a compositional theory of tune interpretation, according to which the pitch accents H^* and L^* are meaningful themselves, i.e., directly representing meanings like newness, salience, mutual belief, incredulity, etc. Interestingly, this again differs from the classic assumption that segmental phonemes like consonants and vowels are not directly meaningful, but serve only as building blocks of morphemes that are the smallest units of meaning.

In general, therefore, as explained by Shattuck-Hufnagel & Turk (1996), analysis-by-transcription is not meant to develop predictive knowledge per se, but only as an initial step in that direction. But because the transcription systems themselves are not developed from empirical data, the lack of reference problem mentioned above is not fully addressed. In more recent research, analysis-by-transcription is often incorporated into empirical studies, in which the transcriptions are used as measurements and subjected to statistical analysis (e.g., Caspers, 2003; Grice et al., 2009; Mady & Kleber, 2010; Metusalem & Ito, 2008; Yoon, 2010). This seems to be a welcome step, but it may also be beneficial to combine the transcription analysis with other measurements.

3 Analysis by introspection

Analysis-by-introspection refers to assignment of prosodic categories to text based mainly on intuition, without experimental investigation or transcription of recorded utterances. This kind of approach is adopted mainly in theoretical studies of syntax-prosody or pragmatics-prosody interfaces, for example, Chaffe (1974, 1976), Brazil et al. (1980) and Buring (2006, 2007). An apparent issue with this kind of approach is that, as mentioned in the introduction, human introspection is not known to be highly reliable. Thus the assigned prosodic categories could well have been affected by the imprecision of the introspection. Note that from the perspective of hypothesis testing to be discussed next, there is nothing wrong with using introspection as a means of hypothesis formation. What is crucial is not to end there, and treat the introspectively derived categories as established ones.

4 Analysis by hypothesis testing — Experimental approaches

By a broad definition, virtually any research effort contains some experimental elements. This is true of even analysis by introspection. What distinguishes different approaches is how many of the essential elements of a typical experimental paradigm are adopted. In analysis by introspection, the process remains at the stage of hypothesis formation, and in many cases, further hypotheses are proposed on top of the initial hypothesis which itself has not yet been empirically tested. A more thorough experimental approach would consist of not only the formation of a general hypothesis, but also predictions derived from the hypothesis that can be tested against empirical data, thus possibly falsifying the hypothesis. Falsifiability is the hallmark of genuine scientific inquiry according to the Popperian view of science (Popper, 1959). The process of falsification, though highly discipline-dependent, requires many essential components that are by now widely accepted, and have mostly been built into various statistical models. In general, it is essential to be able to independently control all the factors that might potentially confound the effect of the factors being tested. Such control is imperative in both data collection and data analysis. Note that, unlike what is often perceived, experimental control is not about dictating what the subjects should do. Rather, it is about how to guarantee systematic identification and separation of the factors that may significantly contribute to what is under observation. Such guarantee needs to be achieved by clear delimitation of independent, dependent, and to-be-controlled variables in the study design, preferably in a form that permits the application of statistical analysis.

4.1 Functionality versus encoding

For research on prosody, an additional issue, as mentioned in the introduction, is how well a study manages to separate functionality from encoding. This issue is methodologically essential because it is directly relevant for the question of control. Ideally, a prosodic function should be defined in terms of both the kind of information it conveys and its prosodic forms of encoding, but it is vitally important that the two parts of the definition be clearly separated. On the one hand, one would look for prosodic functions that make contrasts that are defined in meaning rather than in form. On other hand, each operational function should have, by definition, forms that directly encode the contrast. So, it is an empirical question whether each hypothesized

function actually has a unique encoding scheme in a particular language. [Table 1](#) (p. 110-111) shows a list of plausible functions and their acoustic correlates as reported by various studies.

As can be seen in [Table 1](#) (p. 110-111), focus and boundary marking have both attracted a large amount of research effort, and clear acoustic cues have been identified. For focus, increasingly clear evidence shows that its cue is not only in the focused items themselves, but also in the post-focal components, which typically show reduced pitch range and intensity. But such *post-focus compression* (PFC) is found only in some of the languages, while many other languages show no evidence of PFC. The cause of such uneven distribution is currently under active investigation (cf. Xu, 2011 for a brief overview). Also related to focus are *prominence* and *newness*, two potential functions that have been widely recognized. Prominence has sometimes been viewed as synonymous to focus, but often it is assumed to be free of any specific linguistic functions. Such freedom, however, leaves room for confounding between better defined functions such as lexical stress and focus. The problem resulting from such confounding is highlighted by the recent controversial finding by Kochanski et al. (2005) that F_0 contributes little to prominence. In the case of newness/givenness, conceptually the contrast can be easily defined (Bock & Mazzella, 1983). However, when examined with other functions systematically controlled, it does not seem to show distinct prosodic correlates independent of focus and topic, other than a small durational effect, at least in Mandarin (Wang & Xu, in press). In the studies on other languages, newness is not genuinely separated from focus and therefore it is yet unknown whether it has unique cues of its own. Another focus-related issue is the different types of focus that have been widely assumed (Gussenhoven, 2007). Of the four studies shown in [Table 1](#) (p. 110-111), that have looked into different types of focus, only one has reported difference between narrow and contrastive focus (Avesani & Vayra, 2003, for Florence Italian). More research is therefore needed to further investigate whether there is distinct prosodic coding for different types of focus.

There have been remarkably many studies looking into issues related to boundary marking, and the list shown in [Table 1](#) (p. 110-111) is far from complete. Overall, the most consistent cue is in terms of duration, whether it is that of domain-final vowel, syllable or word, pause or domain-initial consonant. The emerging picture is that such duration-based boundary marking is applied across the board, from within a word to within a phrase to between phrases to between sentences, as can be seen in Table 2. Also shown in Table 2 is an increase in the number of cues as the units separated by the boundary become larger. Interestingly, as can be seen in the bottom two rows, when the unit is the sentence or larger, boundary cues seem to be joined by topic cues. This may suggest that the topic function could be a special case of the boundary function. This possibility can be explored in future research.

Table 2: Prosodic boundary cues reported in a number of studies. An increase in the number of cues can be seen as the boundary level becomes higher.

Boundary type	Prosodic cues
Within phrase	Duration (Gussenhoven & Rietveld 1992, Nakatani et al. 1981, White & Turk 2010, Xu & Wang 2009)
Between phrases	Duration, F_0 (Edwards et al. 1991, Gussenhoven & Rietveld 1992, Klatt 1975, Nakatani et al. 1981, Wagner 2005)
Between sentences	Duration, pause, F_0 (Berkovits 1993, 1994, Ladd 1988, Swerts 1997, Wang & Xu in press)
Between paragraphs (topic?)	Duration, pause, F_0 (Nakajima & Allen 1993, Lehiste 1975, Swerts 1997)

Within the topic function, there is a subtype known as contrastive topic, which is widely recognized among the syntacticians (Büring, 2003; Kiss, 2008; Krifka, 2008). It is assumed that contrastive topic is associated with a rising pitch pattern known as B-accent (Bolinger, 1965). However, judging from the pitch track provided in Büring (2003) which is cited from Jackendoff (1972), there is no clear evidence of rising F_0 . Also in the course of conducting this review, no systematic empirical data were found on contrastive topic. According to Büring (2007), the contrastive component of contrastive topic is similar to focus. At least for Mandarin, Wang & Xu (in press) found that contrastiveness in topic has no prosodic correlates independent of focus when both topic and focus are controlled. This issue nevertheless needs more investigation for other languages.

The prosody-syntax row in [Table 1](#) (p. 110-111) shows studies on how prosody and syntax influence each other and those on how prosody may aid comprehension in speech perception. The prosodic cues are rather diverse and often not clearly stated. The relation between prosody and syntax is nevertheless a very important issue, and there is a question that has rarely been asked before: Why should the two be matched or linked in the first place? That is, assuming that speech is about exchanging information, if a meaning is already syntactically or prosodically conveyed, why should it be also encoded in the other domain? Unless, of course, *redundancy of coding* is favored (Assmann & Summerfield, 2004). This possibility suggests that exploring cross-domain relation in terms of redundant coding is a direction worth pursuing in future research.

The last two rows of [Table 1](#) (p. 110-111) show that there have been many studies looking at emotions and attitudes. A multitude of cues have been reported for various emotions and attitudes and they are too diverse to be discussed in detail in this review. Interested readers may find a number of detailed literature reviews on vocal expression of emotions and attitudes, including Mauss and Robinson (2009), Murray and Arnott (1993), Scherer (2003) and Scherer and Bänziger (2004). In addition, a new line of research based on evolutionary mechanisms and cross-species comparisons has shown some interesting results (Chuenwattanapranithi et al. 1996; Morton, 1977; Ohala 1984, 1996; Xu & Kelly, 2010; Xu et al., forthcoming).

4.2 Production vs. perception

Among the experimental approaches, there is often a preference for either a production- or perception-oriented strategy. From a functional perspective, an operable function by definition must be contrastively encoded through production, and reliably decoded through perception. The production and perception of a function therefore must have evolved in tandem: only patterns that are perceptually distinct and articulatorily possible could have emerged and been maintained. It is therefore beneficial for both perception- and production-oriented studies to take a comprehensive view in their interpretation of the empirical data. In the following, I will highlight a few issues as an illustration of the importance of a comprehensive view.

Pitch contour stylization is a strategy to find a piecewise linear approximation of the F_0 curve that is perceptually indistinct from the original contours. It was developed in the 60-80s with the assumption that pitch movements that are interpreted as relevant by listeners in non-linguistic perceptual judgment tasks directly reflect the intention of the speaker ('t Hart et al. 1990). An important issue about this and other stylization approaches is generalizability. While piecewise linear approximations may be good enough for specific utterances, it may not be applicable to other utterances with different segmental and lexical compositions. This is because linear approximations have no built-in mechanisms to handle contextual variations, which alter F_0 height, movement direction and alignment of turning points (Gandour et al., 1994; Potisuk et

al., 1996; Xu 1999). These contextual variations have many different sources, as summarized in Xu (2006), and each need to be handled in terms of its own specific underlying mechanism.

Likewise, for any pattern found in production, there is also the question of perceptual relevance and sensitivity. A case in point is the many acoustic patterns found to be associated with focus in production studies. There has been some research on the perceptual relevance of focus-related pitch patterns (Botinis, 2000; Prom-on et al. 2009; Rump & Collier, 1996; Xu et al., 2004). Prom-on et al. (2009) also found that incorporating focus-appropriate duration changes further improves focus identification rate based on F_0 only changes. However, there has been virtually no research examining the perceptual relevance of intensity (Kochanski et al. 2005) and spectral balance (Sluijter & van Heuven, 1996).

Another pattern found in recent research is a rather consistent temporal alignment of F_0 turning points relative to segmental events, especially to syllable edges (Arvaniti & Garding, 2007; Atterer & Ladd, 2004; Dilley et al., 2005; D'Imperio et al., 2007; Gili Fivela, 2002; Prieto, 2009; Prieto & Torreira, 2007; Shue et al., 2010; Xu, 1998, 1999, 2001; Xu & Xu, 2005). The alignment has been found to be also sensitive to accent type, vowel length, syllable structure and focus position, as well as being language- or even dialect-dependent (Arvaniti & Garding, 2007; Atterer & Ladd, 2004; D'Imperio et al., 2007; Leyden & Heuven, 2006). The perceptual significance of such alignment is only beginning to be investigated (Dilley, 2005; Niebuhr, 2007; Tokuma & Xu, 2009, 2011). Niebuhr (2007) found that listeners are sensitive not only to turning-point alignment, but also to changes in the shape of F_0 trajectories *due to the alignment change*. This means that information is not necessarily encoded in terms of F_0 peak timing. Consistent with this possibility is the view that the actual encoding units are the underlying pitch targets that are always synchronized with the syllable, and the alignment pattern is the result of interaction between the underlying targets and other factors such as surrounding targets, stress, position of the syllable, focus and duration, etc. (Xu, 2005; Xu & Wang, 2001). Evidence for target-syllable synchronization and target-context interaction can be seen in Dilley (2005), Gao & Xu (2010), Lu and Xu (2006, 2007), Niebuhr (2007) and Xu (1998, 2001). Nevertheless, more research is needed to further resolve this issue by examining the relation between F_0 alignment, articulatory mechanisms and perceptual sensitivity.

4.3 Ecological validity

An issue that faces most, if not all, of experimental studies is the question of ecological validity. That is, how much of what is observed in a controlled experiment is applicable to everyday speech. To address this question, we need first to clarify what is meant by applicability. When made explicit, it could mean one of several things: a) relevance for explaining the prosody of utterances in real life, b) relevance for processing the prosody of real-life utterances in speech recognition, c) relevance for generating naturalistic prosody in speech synthesis, and d) relevance for other practical purposes such as language teaching and speech therapy. For any of these tasks, it would be crucial to have *predictive knowledge* about prosody. So, the real question about ecological validity of a study should be, can it lead to knowledge that is useful for the explanation, recognition, synthesis or other applications of prosody?

Experimental investigations, by systematically controlling various factors, are designed to develop predictive knowledge, i.e., knowledge that is generalizable to other similar situations.¹

¹ Predictive knowledge can be also viewed as the kind that allows us to launch a rocket into the air, knowing that it will reach the moon rather than falling into the ocean, or to give a medicine to a patient, knowing that it will cure rather than kill him. In speech science, predictive knowledge would be

However, also because of the need for systematic controls, individual studies cannot examine all factors at once, and so each is necessarily limited in scope. For example, in a typical lab experiment, subjects are not asked to produce sentences with strong emotions, unless the study is about emotional expressions. Are the findings, then, still valid in cases where emotion is involved? This kind of question should also be empirically answered. There is some initial evidence that functions like focus can be encoded in parallel with emotions (Xu et al., forthcoming). Bruce and Touati (1992) have demonstrated that prosodic patterns found in read speech in Swedish are also found in spontaneous speech, in which emotions are certainly involved (political debate, radio listener call-in conversation). More such research is needed to further test the applicability of experimental findings to spontaneous speech.

Likewise, studies that intend to directly embrace the richness of natural, spontaneous prosody should also go beyond only developing descriptive knowledge, and aim also at establishing predictive knowledge that can be applied in the explanation, recognition, synthesis and other applications of natural prosody.

4.4 Analysis of spontaneous speech

One possible way to improve ecological validity and capture the rich amount of information carried by prosody is to record spontaneous speech in a natural environment. In this strategy, samples of speech are recorded from spontaneous conversation either face to face (Nakajima & Allen, 1993; Swerts, 1992) or over the phone (Campbell, 2001; Cieri & Liberman, 2006), in the classroom (Tseng et al., 2010) or under other naturalistic conditions (Bruce & Touati, 1992; House, 2005; Tseng et al., 2010). One of the most elaborate efforts is the JST CREST ESP Project in which hundreds of hours of spontaneous conversations were recorded, some of which were real-life daily conversations (Campbell, 2000, 2001, 2004). The potential advantage of spontaneous speech over so-called lab speech is that it allows the observation of various phenomena that are lacking in the latter, including discourse markers (Hirschberg & Litman, 1987; Hansson, 1999), interactive structure (Edlund & Heldner, 2005; Oliveira & Freitas, 2008; Swerts & Ostendorf, 1997), turn taking (Caspers, 2003; de Ruiter et al., 2006; Oliveira & Freitas, 2008; Schafer, 1984), repair (van Wijk & Kempen, 1987), filled pauses (Shriberg & Lickley, 1993) and emotion and attitude (Campbell, 2004).

There are various difficulties in studying spontaneous speech, many of which are discussed in detail by Beckman (1997). In particular, its effectiveness depends much on the method of data collection, just as in more controlled studies discussed above. As cautioned by Beckman (1997:19), “the researcher must carefully attend to many aspects of the elicitation paradigm in order to have any luck in getting spontaneous speech that will be useful for the research purpose.” From the perspective of separating functionality from encoding, spontaneous speech presents compounded difficulty. An inherent stumbling block is that, exactly because spontaneous speech is rich in prosodic functionality, each function needs to be properly labeled. Otherwise what is observed for any particular function would easily be confounded by other

the kind that will allow us to build a robot that sounds like a real human, saying all the right things with all the right tone of voice, or a robot that will understand every word we are saying, and every subtlety in our tone of voice, or at least to the extent we are able to do ourselves, or to apply a teaching method to a language learner, knowing that it will work better than chance, and better than any potential “placebo” effect. There is no doubt that we are still a long way away from such predictive knowledge, and most of the research is not directly concerned with practical issues. But the lessons from other scientific disciplines are that predictive knowledge developed in basic research is the key to real advances in technology.

functions. As explained by Campbell (2004:300): "... each utterance must be evaluated separately for such features as the relationships between speaker and hearer (age, sex, familiarity, rank, politeness, etc.), the degree of commitment to the content of the utterance (citing, recalling, revealing, acting, informing, insisting, etc.), the long-term moods and short-term emotions and the attitudinal states of the speaker, the pragmatic force behind the speech act, the voice-quality underlying the utterance (breathy, relaxed, pressed, forced), and so on." If any of these factors is not properly labeled, there is a danger that other factors cannot be properly recognized either. A brief summary in [Table 3](#) (p. 112) demonstrates the diversity of annotations done in a number of studies on spontaneous speech.

A close examination of the studies shown in [Table 3](#) (p. 112) shows that they are mostly focused on discourse-related functions, including turn taking, boundary marking, filled pauses and self-repairs. Also examined are emotions, speaking style and pitch accent types during discourse. Most of these phenomena, with the exception of marking boundaries of lower strengths and certain types of accents, are not easily elicited under laboratory conditions. They are therefore more suitable for this type of studies.

After the collection of the corpus, various analyses need to be conducted. There are almost endless possible ways to analyze a corpus, but the exact method used would depend on the purpose. Similar to the problem of labeling, the potential challenge in the analysis of spontaneous speech is how to overcome confounding of different factors. Various strategies have been developed. Swerts (1997), for example, used listener perception to first rank the boundary strengths, and then examine the differential cues involved in boundaries of different strengths. Nakajima and Allen (1993) examined prosodic events in map task dialogues. Instead of examining them in terms of turn taking, they classified pitch events in terms of topic shift, continuation, elaboration, etc., which seems to correspond to pitch shift of different sizes across the turns. More methodological innovations in future research should further help overcome the intrinsic difficulties with spontaneous speech.

5 Analysis by Modeling

Potentially the most rigorous test of our understanding of prosody, especially in terms of predictive knowledge, is computational modeling, which, though also one type of experimental approach, is discussed separately because its importance has in general not been duly recognized. In modeling, attention to detail can be pushed to the limit, because our knowledge is checked against all the minutiae of reality. To computationally model prosody, quantitative algorithms need to be developed to generate continuous prosodic events whose every detail can be compared to that of real speech. The development of each algorithm is based on a particular understanding of or assumption about an underlying mechanism of prosody. Of course, the rigor of computational modeling as a research tool is not automatically guaranteed, and it is in fact more often not fully realized. This is because its effectiveness depends on a number of critical aspects of the modeling process, including, in particular, (a) description vs. prediction, (b) method of evaluation, and (c) degree of freedom and level of control.

5.1 Descriptive vs. predictive modeling

If the goal of modeling prosody is only to find close mathematical representations of intonation patterns of individual utterances, the most straightforward way is probably to use polynomial functions of various types (simple, spline, piecewise linear, etc.) to fit the F_0 contour of each

utterance. This has been done in a number of studies for tone and intonation (Andruski & Costello, 2004; Chen & Chang, 1992; de Ruiter, 2008; Gandour et al., 1999; Grab et al., 2007; Hirst et al., 2000; Liu et al., 2006). With polynomial fitting, each section of a complex contour can be represented by a set of coefficients, which may drastically reduce the amount of representations from the original point-by-point data. A critical question about polynomial representations, however, is whether they are linguistically meaningful and whether they can be used in predictive modeling, i.e., serving as categorical parameters that can be generalized to other instances of the same category. In the aforementioned studies, the polynomial coefficients are used only in statistic comparisons or classifications, but not in predictive synthesis. So, the predictiveness of models based on polynomial fitting is still unknown. Similarly, a number of other studies that use non-polynomial models also only fitted the F_0 contours of specific utterances without predicting new F_0 contours (Anderson et al., 1984; Bellegarda et al., 2001; Fujisaki, 1983, 1988, 1992; Fujisaki et al., 1994, 2003; Mixdorff & Fujisaki, 1997, 2000; Pierrehumbert, 1981; Taylor, 2000a; van Santen et al., 2005; Veronis et al., 1998). When the modeling experiment is non-predictive, it is hard to know how much of the abstract characteristics of each prosodic category have been captured.

Whether for the sake of building a practical synthesis system or increasing knowledge about prosody, therefore, it is imperative to develop models that are predictive. Predictiveness, however, can be achieved at different levels. The highest degree of predictiveness would be found in a system with human-like performance, i.e., starting from idea formation and finishing with production of fully natural and informative prosody. It will probably be a long time before anything close to that is developed, of course. The next best would be something akin to a concept-to-speech system (McKeown & Pan 2000; Taylor, 2000b; Young & Fallside, 1979), which, though also very tantalizing, seems to be still far from materialization. Part of the difficulty of concept-to-speech is the translation of concepts into proper functional prosodic categories. But this is also the problem facing systems designed to achieve predictability at an even lower level. That is, given a set of utterances that are functionally marked, regardless of whether the category labels are derived from text or concepts, or determined by human labelers, can the system then generate prosodic forms that fit closely to those of the original? From the perspective of analysis by modeling, performing such prediction is actually a process of testing both the validity of the predictors and the assumed encoding mechanisms.

[Table 4](#) (p. 113-114) shows a summary of the predictors (also known as input features) tested in various modeling studies. Note that some of the studies in the table have used a large number of predictors, e.g., Bellegarda et al. (2001), Mixdorff and Jokisch (2001), Mohler and Conkie (1998) and Sun (2002). In these cases, for the sake of developing predictive knowledge, it would be desirable to not only assess the overall quality of data fitting, but also examine the contribution of each predictor. Note also that the performance of a particular predictor does not necessarily reflect its importance, because the performance should also be closely related to its modeling implementation based on the assumed encoding strategy. As a tool for hypothesis testing, prosodic modeling therefore should be a continuous process of not only trying to identify all the relevant predictors, but also searching for the proper encoding mechanisms, which are not necessarily shared across the predictors (cf. Xu, 2005 for some hypotheses).

5.2 Method of evaluation

Whether a model is descriptive or predictive, its effectiveness needs to be evaluated one way or another. The methods of evaluation have been quite diverse, ranging from informal visual inspection and listening to various formal objective and subjective evaluations. The goal of

objective evaluations is to measure goodness of fit between synthetic and original prosodic forms. There have been three major methods: Root Mean Square Error (RMSE) (Fujisaki et al., 2005; Kochanski et al., 2003; Ni et al. 2006; Prom-on et al., 2009; Raidt et al., 2004; Sakurai et al., 2003; Sun, 2006) and Pearson's correlation or correlation for short (Mixdorff & Jokisch 2001; Prom-on et al., 2009; Raidt et al., 2004; Sun, 2006), and mean absolute frequency deviation across each utterance (Bellegarda et al., 2001). Among the three, RMSE and correlation are the most used. Hermes (1998) has shown that both measures are effective. For RMSE, the unit of measurement varies from Hz (Mohler & Conkie, 1998; Morlect et al., 2001; Sun, 2002) to semitone (Prom-on et al., 2009; Raidt et al., 2004) to $\ln(F_0)$ (Fujisaki et al., 2005; Gu et al., 2006; Ni et al., 2006; Sakurai et al., 2003), which makes cross-study comparison difficult. Although $\ln(F_0)$ and semitones can be mutually converted from one to the other because both are logarithmic ($\text{semitone}_{\text{RMSE}} = 12 \ln(F_0)_{\text{RMSE}} / \ln(2)$), RMSE values in Hz cannot be properly converted to a logarithmic scale, because of lack of reference, see equation 1. The advantage of logarithmic scale for intonation has been shown in terms of both production (Fujisaki, 2003; Nolan, 2003; Xu & Sun, 2002) and perception (Traunmüller & Eriksson, 1994). Given such logarithmic nature, RMSE values in Hz need to be interpreted differently depending on the average pitch of the speaker. For example, an RMSE value of 10 Hz (≈ 1.39 st) for a male speaker with a mean F_0 of $F_{0\text{reference}} = 120$ Hz is roughly equivalent to that of 18 Hz (≈ 1.36 st) for a female voice with a mean F_0 of $F_{0\text{reference}} = 220$ Hz.

$$\text{semitone} = 12 \log_2(F_0 / F_{0\text{reference}}) \quad (1)$$

In addition to the objective evaluations, the effectiveness of a model can also be perceptually evaluated. The perceptual evaluations fall into two major types. In the first type, the naturalness of the synthetic prosody is evaluated. Various methods have been used, including direct judgment of naturalness (Sun, 2002), judgment of whether the prosody is synthetic or natural (Prom-on et al., 2009), and similarity rating (Ni et al., 2006). Some studies (e.g., Mohler & Conkie, 1998) have made use of mean opinion score (MOS) a measurement designed for evaluating audio media quality, which employs a 5-level scale:

Mean opinion score (MOS)		
MOS	Quality	Impairment
5	Excellent	Imperceptible
4	Good	Perceptible but not annoying
3	Fair	Slightly annoying
2	Poor	Annoying
1	Bad	Very annoying

The MOS is then calculated as the arithmetic mean of all the individual scores. This method seems to have the potential of standardizing formal naturalness evaluations. But its advantage has yet to be fully demonstrated.

In the second type of evaluation, what is assessed is the perceptual accuracy of the hypothetical categories being modeled, such as lexical tone, lexical stress, pitch accent, focus, boundary strength, sentence modality, type of emotion and attitude, etc. Surprisingly, only two of the studies reviewed here performed this kind of evaluations (Morlec et al., 2001; Prom-on et al., 2009). All the other studies that performed perceptual evaluations only assessed perceptual naturalness of synthetic prosody. For the sake of improving the predictiveness of the models and

using them as effective hypothesis testing tools, it is highly desirable in future research to include perceptual identification of the input categories as one of the evaluation methods.

5.3 Degrees of freedom and Level of control

Degrees of freedom (df) refers to the number of free parameters needed to completely specify a model. These are the parameters that must be estimated in order for the model to generate the intended output. Degrees of freedom is seldom discussed, however, and in many theories and models the number of free parameters is kept implicit. The importance of degrees of freedom is that it reflects assumptions about issues such as what is speaker-controlled and what is articulatorily mandatory, what is functionally encoded and what is perceptually relevant. Making these assumptions explicit will allow various tradeoffs related to degrees of freedom to be directly evaluated. The following are some common issues closely related to degrees of freedom.

- 1) What are the assumed basic prosodic events and how are they specified? For example, in an increasing order of df, the assumed basic events can be single points, straight lines, piece-wise linear shapes or complex contours.
- 2) Are the properties of the basic events fixed or variable? Fixed properties often entail lower df than variable properties. For variable properties, how many parameters are needed to specify the variability?
- 3) How is the timing of the basic events specified? The df would be larger if the timing is freely variable and thus needs to be estimated than if the timing is fixed relative to segmental events such as the onset, offset or the entire interval of vowel, rhyme, syllable or word.
- 4) Are transitions needed between the basic events? If yes, how many parameters are needed to specify each transition?
- 5) Are global parameters, such as top and base lines, needed to specify each local contour?

[Table 5](#) (p. 115) shows a comparison of degrees of freedom in a number of models for controlling local F_0 contours. What is shown here, however, is only degrees of freedom for generating local pitch events. Note that even at this level, degrees of freedom should not be used as the sole criterion for judging a model, and it is not the case that smaller degrees of freedom is always better. What is more crucial is whether each degree of freedom has sufficient justifications. Ideally the degrees of freedom should reflect the assumed human control, at least at the level that the model tries to simulate. For example, if transitions between basic events are specified by one or more free parameters, it would be helpful to explicate the articulatory nature of such controlled transitions. Furthermore, degrees of freedom also has to do with the level of control the model tries to simulate, e.g., at the level of the muscle (Fujisaki et al., 2005), the underlying target and its articulatory approximation (Kochanski & Shih, 2003; Prom-on et al., 2009), the decomposed holistic contours (Baily et al., 2005) or surface contours (Pierrehumber, 1981; Taylor 2000; van Santen et al. 2005).

For more global events related to higher functions as shown in [Table 1](#) (p. 110-111), more degrees of freedom is certainly needed. For each contrastive prosodic component above the basic units, at least a single degree of freedom is required to simulate its coding. Not implementing this degree of freedom would result in a decrease in the functionality of the model.

6 Concluding remarks: Need for linking and integration

We have seen through this brief review that much effort has been made to build up a body of predictive knowledge on speech prosody. The goal of this review has been to highlight the differences between the various methodological approaches in terms of their effectiveness in solving the lack of reference problem. We have seen that there is mounting evidence that focus, boundary marking and modality are three highly likely functions, and there seems to be converging evidence for their distinctive prosodic coding. The picture for other functions is less clear. At the same time, we can also see a striking disconnect between subareas of prosody research. For example, although I have used the term analysis by modeling, the truth is that most of modeling activities are not conducted with the aim to test any existing theory or to develop a new theory, but are rather done for modeling's own sake. As a result, major theories of prosody have yet to be put through the most rigorously test, namely, being forced to make predictions about all the prosodic details that can be checked against real speech, or to recognize prosodic functions from real speech. Similar disconnect, though to a lesser degree, can be also seen between perception-oriented efforts and production-oriented ones. It is therefore highly desirable for there to be much more linking between the subareas and different approaches and hopefully also true integrations. If such linking and integration are to occur, the next ten years should see some truly accelerated advances in both theoretical development and practical applications.

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Table 1: Hypothetical prosodic functions and their reported encoding schemes. [\[back\]](#)

Functions	Languages	Main acoustic correlates	Studies
Focus	Egyptian Arabic, English, French, Hindi, Irish, Japanese, Italian, Korean, Lebanese Arabic, Mandarin, Persian, Swedish, Tibetan, Turkish, Uygur	On-focus increase of pitch range, duration and intensity; post-focus compression and lowering of pitch range and intensity; change of pitch targets of on-focus stressed syllable (English)	Bagshaw 1993, Beyssade et al., 2009, Bolinger 1961, Botinis et al. 1999, Botinis et al. 2000, Bruce 1982a, 1982b, Chahal 2003, Chen 2006, Chen & Destruel, 2010, Chen et al., 2009, Cooper et al., 1985, de Jong 1995, 2004, de Jong & Zawaydeh 2002, Delais-Roussarie et al. 2002, Di Cristo A and Jankowski 1999, D'Imperio 2001, Dohen & Lævenbruck 2004, Dorn & Ni Chasaide, Eady et al., 1986, Gerard & Dahan 1995, Face 2006, Féry & Kügler 2008, Heldner 2003, Heldner & Strangert 2001, Hellmuth 2006, Ipek 2011, Jin 1996, Kabagema-Bilan 2011, Lecumberri 1997, Lee 1956, Lee & Xu 2010, Nootboom & Krut 1987, Patil et al. 2008, Rump & Collier 1996, Sadat-Tehrani 2009, Sugahara 2002, 2005, Ueyama & Jun 1998, Wang et al. 2011, Xu 1999, Xu & Xu 2005, Xu et al. 2004,
	Cantonese, Buli, Chichewa, Chichewa, Chitumbuka, Dagbani, Deang, Durban Zulu, Ewe, Hausa, Konni, Northern Sotho, Qiang, Taiwanese, Tangale, Vietnamese, Wa, Wolof, Yi, Yucatec Maya	<i>Lack of post-focus compression of pitch range and intensity</i>	Chen et al. 2009; Downing 2008, Gu & Lee 2007; Hartmann 2008, Hartmann & Zimmermann 2007, Jannedy 2007, Kügler & Skopeteas 2007, Nguy et al. 2008, Pan 2007, Rialland & Robert 2001, Schwarz 2009, Wang et al. 2011, Wu & Chung 2011, Wu & Xu 2010, Xu 2011, Zerbian et al. 2010
Different types of focus	Dutch, German, English, Italian	<i>No consistent differences found between narrow and corrective focus in Dutch, and between narrow and contrastive focus in English</i>	Avesani & Vayra 2003, Baumann et al. 2007, Hanssen et al. 2008, Sityaev & House 2002
Prominence	Dutch, English	Similar to on-focus cues; but no explicit examination of post-focus cues	Calhoun, 2010, Gussenhove et al. 1997, Hermes & Rump 1994, Kochanski et al. 2005, Ladd et al. 1994, Rietveld & Gussenhoven 1985, Rump & Hermes 1996, Terken 1991, Terken & Hermes 2000
Newness vs. givenness	Dutch, English, German, Mandarin	<i>Lack of consistent acoustic correlates independent of focus</i>	Féry & Kügler 2008, Most & Saltz 1979, Nootboom & Krut 1987, Wang & Xu in press
Boundary marking, Grouping, structuring	Dutch, English, French, Hebrew, Korean, Mandarin, Brazilian Portuguese	Domain/group-final lengthening, F ₀ reset	Barbosa 2007, Barbosa & Bailly 1994, Beckman & Edwards 1990, Berkovits 1994, Byrd & Saltzman 1998, 2003, Campbell, 1993, Carlson et al., 2001, Cho 2004, Cho & McQueen 2005, Edwards & Beckman 1988, Ferriera 1993, Fougerson 2001, Gussenhoven & Rietveld 1992, Krivokapic 2007, Nakatani et al., 1981, Ouden et al., 2009, Redi & Shattuck-Hufnagel

			2001, Sanderman & Collier 1996, Swerts 1997, Wagner 2005, Wightman et al. 1992, Xu & Wang 2009
Topic	Dutch, English, Mandarin	Increased sentence-initial F_0	Lehiste 1975, Sluijter & Terken 1993, Smith 2004, Tseng 2008, Umeda 1982, Wang & Xu in press
Contrastive topic	—	—	<i>No experimental investigations</i>
Turn taking	Dutch, English	Map Task dialogues	Caspers 2003, de Ruiter 2006, Oliveira & Freitas 2008, Schafer 1983
Modality, question vs. statement	Danish, English, German, Japanese, Russian, Swedish	Descriptive monologues, map task dialogues	Delattre et al. 1965, Eady & Cooper 1986, Gårding 1979, 1987, Ho 1976, 1977, Lin 2004, Liu, 2010, Liu & Xu 2005, 2007a, 2007b, Makarova 2001, McRoberts et al. 1995, Studdert-Kennedy & Hadding 1973, Thorsen 1978, 1980
	Multiple languages in the Niger-Congo, Nilo-Saharan, Afro-Asiatic, Nilo-Saharan and Afro-Asiatic superfamilies	<i>Lack of final rising;</i> Falling pitch contour, sentence-final low vowel, vowel lengthening, and a breathy utterance termination	Rialland 2009
Prosody-Syntactic interaction, Syntactic disambiguation, impact on comprehension	English, Finnish, French, Georgian, German, Italian, Japanese, Mandarin	Multiple cues	Anttila et al. 2010, Barbosa 2007, Braun & Tagliapietra 2010, Birch & Clifton 1995, Bock & Mazzella 1983, Christophe et al. 2003, Cole et al. 2010, Donati & Nespor 2003, Fery et al. 2010, Grassmann & Tomasello 2010, Ishihara 2003, Kjellaard & Speer 1999, Lehiste 1976, Schafer et al. 2000, Shen 1993, Snedeker & Casserly 2010, Speer et al. 1996
Emotion	French, English, German, Thai, Brazilian Portuguese	Multiple cues	Auberge & Cathiard 2003, Barbosa 2009, Bänziger & Scherer 2005, Baum & Nowicki 1998, Beller 2008, Breitenstein 2001, Bulut & Narayanan 2008, Chuenwattanapranithi et al. 2008, Frick 1985, Gobl & Chasaide 2003, Ladd et al. 1985, Mauss & Robinson 2009, Ohala 1996, Pell 2001, Protopapas & Lieberman 1997, Scherer & Bänziger 2004, Shami & Verhelst 2007, Trainor et al. 2000, Wennerstrom 2001, Wildgruber et al. 2005, Williams & Stevens 1972, Xu & Kelly 2010
Attitude	Dutch, English, French, German, Japanese, Swedish	Multiple cues	Ambrazaitis 2005, Chen et al. 2004, House 2005, Morlec et al. 2001, Mozziconacci 2001, Ofuka et al. 2000, O'Shaughnessy & Allen 1983, Uldall 1960, Wichmann 2002

Table 3: Types of spontaneous speech and labeling schemes used in a number of studies. [\[back\]](#)

Study	Language	Type of spontaneous speech	Prosodic transcription/annotation
Bruce & Touati, 1992	Swedish, French	Restricted samples from conversations, interviews, political debates and radio programs	Accentual prominence; phrasing; pitch range; boundary tones; pausing
Campbell, 2004	Japanese	Phone dialogue, daily conversation	Speaker state, speaking style, voice type
Caspers, 2003	Dutch	Map task dialogues	ToDI (http://todi.let.kun.nl/), transition type
Edlund & Heldner, 2005	Swedish	Map task dialogues	Turn types
Grønnum, 2009	Danish	Descriptive monologues, map task dialogues	Pitch relation between stressed and immediate post-tonic syllable; phrasal intonation contour
House, 2005	Swedish	Human-machine dialogues	Presence of final rise and final focal accent
Kohler, 1997	German	Unspecified	Kiel intonation model (KIM) with markers of 10 prosodic domains
Nakajima & Allen, 1993	English	Map task dialogues	Discourse structure markers in terms of topic boundary classes
Ogden & Routarinne, 2005	Finnish	Phone calls and face-to-face conversations	Overlapping talk, pause location & duration
Shriberg & Lickley, 1993	American & British English	Human-machine dialogues	Filled pauses and surrounding f0 values
Swerts & Ostendorf, 1997	English	Human-machine dialogues	Discourse segmentation, utterance purpose
Swerts, 1997	Dutch	Monologues describing paintings	Boundary tones, boundary strength
Terken, 1984	Dutch	Instruction monologues	Pitch accents based on perceptual relevant F ₀ movements
Tseng et al., 2010	Mandarin	Classroom lectures	Discourse boundaries and phrasing units, emphasis
van Wijk & Kempen, 1987	Dutch	Picture description task	Self-repairs types

Table 4: Predictors (input features) used in various studies modeling pitch contours and duration. [\[back\]](#).

Study	Language	Predictors (input features)	Output
Bailly & Holm 2005	French	Prosodic attitudes applied to sentences, dependency relations applied to syntactic constituents of read text or operands/operators of spoken math, cliticization typically applied to determiners and auxiliaries, narrow focus applied to words, lexical tones in Mandarin	F ₀ , duration
Bellegarda et al. 2001	American English	ToBI transcription 40 factors, unspecified	F ₀
Barbosa 2007, 2009	Brazilian Portuguese	Position and magnitude of underlying phrase stress, and a set of dynamical control parameters	Duration
Kochanski et al. 2003	Mandarin	Lexical tone, prosodic strength, position in word	F ₀
Fletcher & McVeigh 1993	Australian English	Number of phonemes, nature of syllabic peak, position in word, position in intermediate & intonational Phrase, degree of stress, grammatical function of word, position in foot	Duration
Fujisaki et al. 2005	Mandarin	Syllable duration, amplitude of preceding tone command, which constrain timing and amplitude of tone command	F ₀
Mixdorff & Jokisch 2001	German	Boundary depth, strength, nucleus schwa/non-schwa, type of intoneme, part-of-speech, phrase index in sentence, number of phones in syllable onset, number of phones in syllable rhyme, duration of preceding phrase, <i>amplitude</i> of preceding phrase command, duration of current phrase, distance from preceding phrase command, coda voiced	F ₀
Mohler & Conkie 1998	American English	43 unspecified features, including accent type	F ₀
Morlec et al. 2001	French	Declarative, question, exclamation, incredulous question, suspicious irony, obviousness, Inter Perceptual Centre Group ratio (IPCG_ratio)	F ₀ , duration
Ni et al. 2006	Mandarin	Tone, sentence modality	F ₀
Prom-on et al. 2009	American English, Mandarin	Tone, lexical stress, focus, position in sentence	F ₀
Raidt et al. 2004	French, German	SFC: Utterance level — modality and prosodic attitude at the utterance level; syntactic structure mathematical operators IGM: 14 input parameters: Syllable level — nature of the phones included in the components of the syllable (syllable, onset, rhyme); word level — accentuation and the part of speech; phrase level or higher — neighboring boundaries and composition of current unit	F ₀
Sakurai et al. 2003	Japanese	Position of accentual phrase in utterance, number of morae, accent type, number of words, part of speech of first and last words, conjugation of first and last words	F ₀
Sanderman & Collier 1996	Dutch	Perceived boundary strength	Duration, stylized F ₀ contour

Sun 2002	American English	Vowel type, coda type, syllable stress, syllable position in word, number of syllable in word, pitch accent type, phrase accent type, number of syllables from major phrase break, part of speech, word position in sentence, number of words from major phrase break	F ₀
van Santen & Shih 2000	American English, Mandarin	No. of phonemes in the syllable, nature of syllabic peak (tense / lax vowel / diphthong / sonorant consonant), Position of syllable in foot, Position of syllable in phrase and clause, Stress assigned to syllable, and nature of pitch movement, Function/content role of the parent word	Duration

Table 5: Degrees of freedom (number of free parameters to be specified) of various models at the local level. [\[back\]](#)

Model	Study	df	Free local parameters
SFC	Bailly & Holm 2005	5	3 F_0 values per vocalic nucleus, 1 lengthening factor, Temporal scope of function
Stem-ML	Kochanski et al. 2003	8	Tone template (5 pitch values), word strength, position of template relative to syllable, length of template relative to the syllable
Fujisaki	Fujisaki et al. 2005	5	Amplitude of 1st command of syllable, amplitude of 2nd command of syllable, onset time of 1st command of syllable, end time of 1st command (and onset of the 2nd command if the second command exists) of syllable, end of 2nd command if 2nd command exists
Tone transformation	Ni & Hirose 2000, Ni et al. 2006	15	F_0 peaks for each tone (2 parameters, bottom and top frequencies of voice register of speaker, bottom and top values of voice register on the RONDO scale, damping ratio of forced vibration, peak coordination (2 parameters), parameters controlling rising characteristics of tone (2), parameters controlling falling characteristics of tone (2)
Sagging transition	Pierrehumbert 1981	4	F_0 topline (height, slope), F_0 baseline (height, slope), height of F_0 turning point (target value), time of F_0 turning point
qTA	Prom-on et al. 2009	3	Target height, target slope, rate of target approximation
Target approximation	Sun 2002	3	Target height, target slope, rate of target approximation
Tilt	Taylor 2000	5	Tilt, tilt amplitude, tile duration, tilt alignment, syllabic position